Ryan Dielhenn

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Technical Skills

Languages Rust, Go, Java, Scala, C, Python, SQL
Systems Docker, Kafka, Prometheus, Git, Linux, CI/CD
Concepts Distributed Systems, LSM-Tree Storage Engines, Object Storage (S3), API Design, High-Performance Data Processing

Projects

SlateDB & OpenData

Open Source Contributor — *February 2026 – Present*

- Designed and implemented distributed compaction for SlateDB (Rust, LSM-tree on object storage), enabling parallel compaction across multiple workers to address a single-compactor throughput bottleneck; implementation merged to main.
- Authored the RFC for the design, covering work scheduling, coordinator/worker split, and failure-handling semantics.
- Contributed to OpenData, a collection of databases sharing a unified object storage foundation via SlateDB; authored and iterated on PRs for enhancements to the TSDB including a regression benchmark, Prometheus metadata endpoint, federate endpoint, and scalar argument support for PromQL.

ZephyrCache

Distributed Systems Project — *July 2025 – May 2026*

- Built a fault-tolerant distributed cache in Go with consistent hashing and replication across ring successors for data durability.
- Exposed a REST API for simple cache and health check operations, along with metrics collection via Prometheus.
- Reviewed and integrated a contributed SWIM-based UDP gossip protocol for decentralized peer registration, failure detection, and membership management as a lightweight alternative to etcd.

Experience

Confluent — *Mountain View, CA*

Software Engineer — *January 2021 – July 2022*

- Contributed to Apache Kafka's migration from ZooKeeper to KRaft, improving observability and reliability of the distributed consensus layer in Java/Scala.
- Built metrics pipelines in Java/Scala to monitor cluster health, quorum state, and inter-broker communication for KRaft, enabling faster incident detection for customer-facing clusters; updated Confluent Cloud tooling in Go to integrate new metrics.
- Contributed to Confluent Cloud's Cluster Linking integration with KRaft architecture for cross-cluster replication.

Confluent — *Mountain View, CA*

Software Engineering Intern — *May 2020 – August 2020*

- Implemented dynamic client reconfiguration in Java/Scala for Apache Kafka, enabling runtime updates to producer/consumer settings (including connection, security, retry, and ack configurations) without service restarts.
- Enhanced Confluent Cloud's rebalance tooling with asynchronous replica movement support.
- Continued contributing to Apache Kafka during Fall 2020 while completing undergraduate degree.

University of San Francisco — *San Francisco, CA*

Undergraduate Researcher — *January 2020 – May 2020*

- Developed a C implementation of Geopresence, a bitmap-based geospatial indexing system using RoaringBitmap compression and HyperLogLog++ for efficient location queries on IoT devices.
- Implemented C-based adaptive grid index achieving 17x speedup over Java; outperformed R-tree indexing by 400x at scale (1M+ points) while R-trees showed advantages on sparse datasets (<7K points).

Education

California State University, Los Angeles — *Los Angeles, CA*

M.S. in Computer Science (In Progress), GPA: 4.0 — *Expected December 2026*

- **Thesis:** Trust and Byzantine Fault Tolerance in Drone Swarms.
- Designing a physics-based trust mechanism that detects Byzantine drones from reported position and movement data, evaluated via event-driven simulation in Java.
- Relevant Coursework: Advanced Computer Networking (BGP, OSPF, TCP/IP), Design and Analysis of Algorithms, Software Engineering, Artificial Intelligence, Machine Learning.

University of San Francisco — *San Francisco, CA*

B.S. in Computer Science, Minor in Mathematics, GPA: 3.75 — *2016 – 2020*

- Relevant CS Coursework: Operating Systems, Computer Architecture, Data Structures & Algorithms, Big Data, Software Development, Programming Language Paradigms, Senior Capstone
- Relevant Math Coursework: Linear Algebra, Formal Methods (Logic and Proof), Calculus I & II, Modern Algebra